EDOARDO ARTURO CANCELLI

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EDUCATION

Politecnico di Torino, Torino, Italy *2024-2027 (expected)*
Aerospace engineering degree (3-year Bachelor's Degree)
Curriculum: General Track (Standard Academic Programme)
Selected for "INTRAPRENDENTI" (top 240 students)

- A 3-year honors program focused on multidisciplinary project-based learning and soft skills.
- Second year choice: Joints: from design to verification (course code : 01HTULZ)

Weighted GPA: 28.68/30

Liceo Scientifico Statale Niccolò Copernico, Udine, Italy *2019-2024*
Liceo Scientifico tradizionale
Track: Potenziamento scientifico informatico
Graduation grade: 90/100

Mount Boucherie Senior Secondary School, Kelowna, Canada *Sep 2022 - Jul 2023*
Exchange student in Canada, British Columbia

CERTIFICATIONS

IELTS Dec 2024 - Dec 2026
Politecnico di Torino *Torino, Italy*

· English language proficiency certification for non-native speakers.
· IELTS 8.0.

EXPERIENCE

Guizzo Space s.r.l Aug 2026 - Sep 2026
Embedded Firmware Engineering Intern *San Vito al Tagliamento, PN, Italy*

· Architected and developed firmware in C for a 24 V, 1.8 A BLDC motor driver on the Microchip ATSAMV71Q21 (Cortex-M7), supporting Field-Oriented Control (FOC), Six-Step commutation, sinusoidal drive, and phase advance.
· Built the FOC chain end-to-end: Clarke/Park transforms, flux and torque current control, Space Vector Modulation (SVPWM) at 20 kHz, and direct drive of the 3 half-bridges.
· Implemented closed-loop speed control using a PID loop, regulating motor speed up to 6 kRPM using the hall-sensor feedback, with electrical angle estimation.
· Integrated current sensing using SAMV71 ADC and Texas Instruments INA for overcurrent protection and torque control.
· Managed version control using Git (GitLab, VS Code), documented codebase with Doxygen and developed across MPLAB X IDE and VS Code.

Guizzo Space s.r.l Aug 2025 - Sep 2025
Embedded Firmware Engineering Intern *San Vito al Tagliamento, PN, Italy*

· Developed firmware in C for a stepper motor driver on the Microchip ATSAMV71Q21 (Cortex-M7), supporting full-step, half-step and micro-stepping.
· Designed a custom sine-wave PWM generator driving the motor through 4 half-bridges.
· Selected the H-bridge via supplier databases (Octopart, Digi-Key, Mouser Electronics).
· Performed debugging and profiling using a Rohde & Schwarz oscilloscope, the Cortex-M7's debug cycle counter and the compiler-generated ARM assembly code.

TECHNICAL STRENGTHS

Computer Languages	C, Python, Java
Tools	Solidworks, Git, GitLab, VS-Code, MPLAB X IDE, MATLAB
Operating Systems	Windows, Ubuntu, Arch Linux

INTERESTS & EXTRACURRICULAR ACTIVITIES

- **Tabletop Gaming, Dungeon Master, D&D:** Developed skills in storytelling, creative thinking, and puzzle design through D&D session design.
- **Motorsports & Aviation:** Interest and research on internal combustion engines, systems, aerodynamics and realistic flight/driving simulations.
- **Hardware Tinkering & Coding:** Experienced in PC troubleshooting, software development, and engineering/optimization games.
- **Mechanical Repairs:** Applied mechanical and electrical skills in repairing and maintaining bikes, household appliances, furniture and custom electronics.
- **Fitness & Winter Sports:** Skiing, snowboarding, strength training.